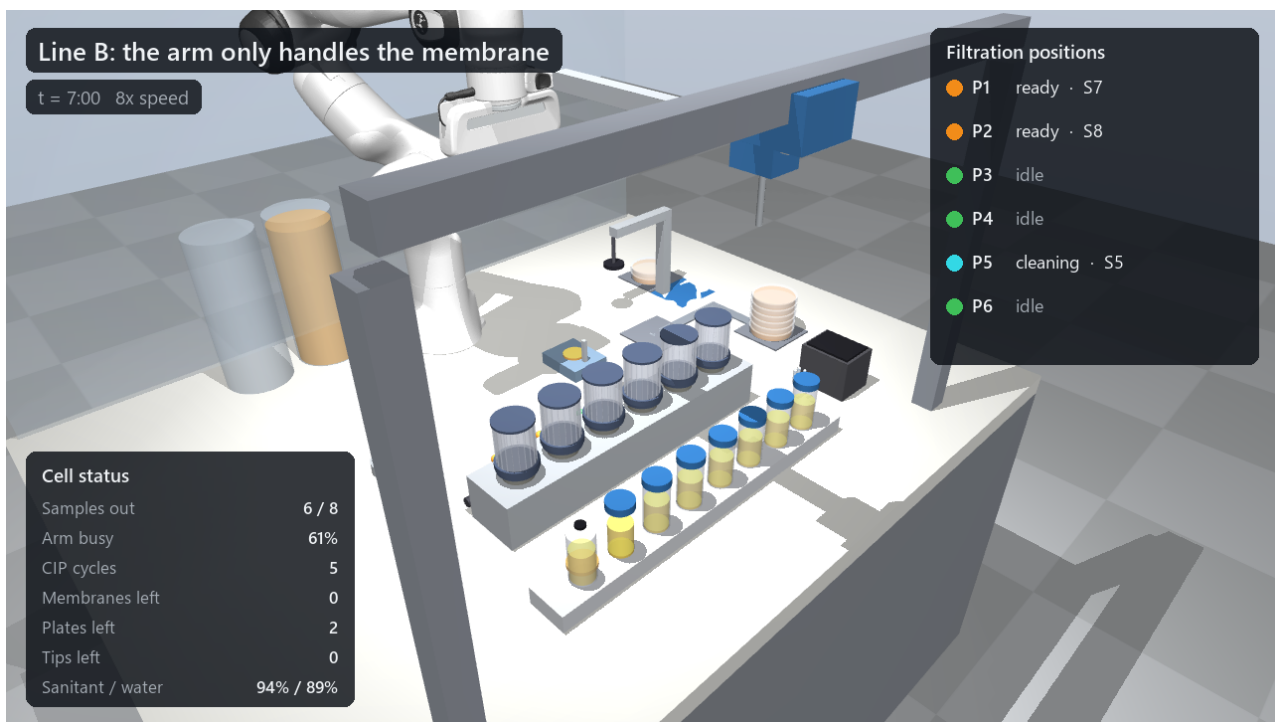


Line B: automating membrane filtration as a process, not a human imitation

Fixed stations for everything routine, the arm only for the wet membrane, with built-in clean-in-place and one-shift material buffers



Question	Answer from the simulation
Throughput	1.0 min per sample (59 samples/h) at 6 positions, against 5 min targeted and 3.2 min for the arm-only cell
Arm workload	37 s of arm time per sample (arm-only cell: 191 s); the rest runs on fixed stations in parallel
Disinfection	clean-in-place on every filtration position, tip wash after every sample, disposable dosing tips; no ethanol or UV
Supply	one operator visit per day at 100 samples/day (membranes 35 h, plates 24 h, tips 23 h)
Status	simulation only; membrane physics, tip design and sanitiser validation need hardware

1 What changed and why

The first simulation automated the process the way a technician performs it: one arm, every step in sequence. It worked (the membrane transfer succeeds in 98% of randomised runs, and the full per-sample sequence runs end to end) but it inherits the technician's productivity: the arm is busy for 191 s per sample, so the cell is limited to about 3.2 minutes per sample however many filtration positions are added. Two further points were raised: the cell must be supplied continuously, and the tools must be disinfected continuously.

The design principle behind Line B is to keep the arm for the one step that needs dexterity, lifting and laying down the wet membrane, and to give every other step to a simple fixed mechanism that can be fast, cleanable and cheap. The arm's cycle is then the only shared resource, and it is short.

Step	Arm-only cell (v1)	Line B
Dosing	arm opens the vessel and pours	gantry doser, disposable tip, through a pierceable septum
Funnel handling	arm carries funnels between frit and wash	6 funnels fixed on the manifold, they lift pneumatically
Funnel cleaning	wash station, ethanol or UV questions	clean-in-place on the position: rinse, sanitant, rinse, air dry
Plates	arm moves plate and lid (69 s per sample)	plate hotel, shuttle and lid lifter, overlapped with filtration
Membrane	arm, from a magazine, twice	arm, from a magazine, twice (unchanged: needs hands)
Tools	not addressed	tweezer tip wash and dry after every sample

2 The cell

Everything below was built and run in the physics simulation (MuJoCo, Franka Panda); fixed mechanisms are animated, the membrane and the arm are simulated with contact. The arm reaches every membrane station; the doser, shuttle and funnels are fixed.

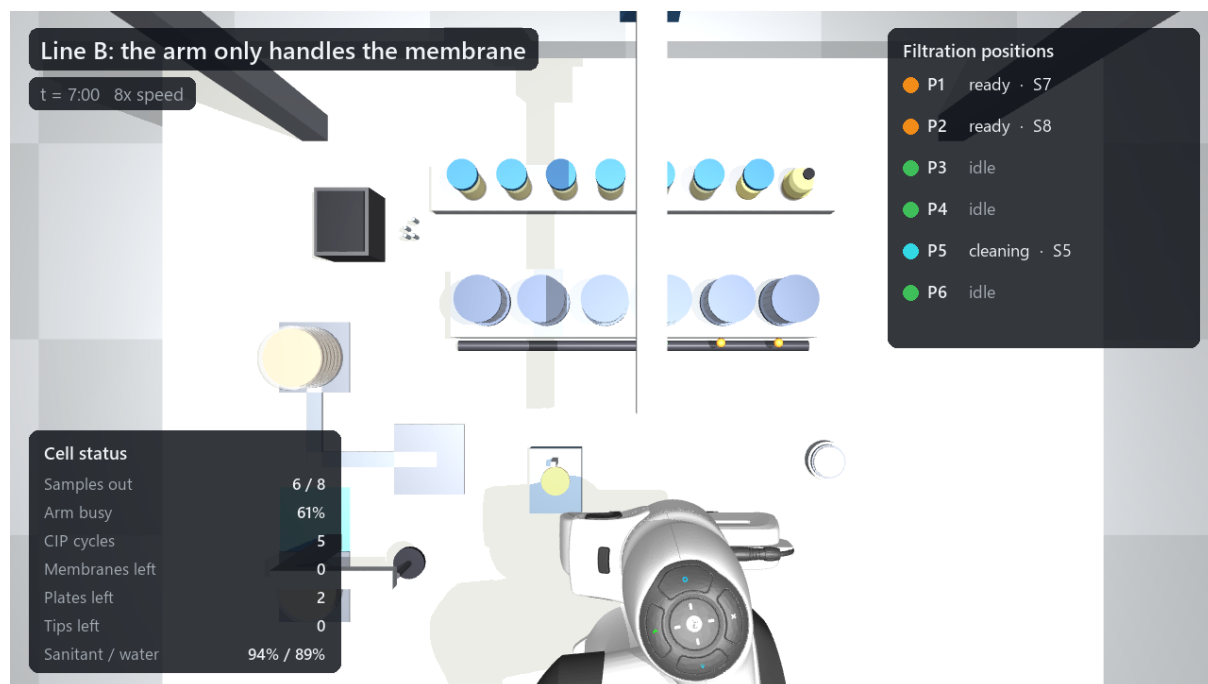


Figure 1. Top view during a run: positions P1 to P6 on the manifold, gantry doser above the vessel rack, plate hotel, shuttle and output stack on the left.

Station	Function	How it is modelled
Membrane magazine	spring-fed stack; the arm picks the top membrane	flex membrane, rigid discs for the stack
Filtration manifold, 6 positions	frit, lifting funnel, valve, status LED	slide-driven funnels; liquid level animated
Gantry doser	Y rail, X slide, Z nozzle; one disposable tip per sample	kinematic, 3 axes, tip rack and waste
Plate hotel, shuttle, lid lifter	stack of closed plates in; open plate at the transfer spot; closed plate out	kinematic, lid and plate follow the carriage
Tip wash	sanitant bath and air knife for the tweezer tips	the arm visits it after each wet transfer
Tanks and hatch	sterile water and sanitant reservoirs; operator hatch	level indicators, counters on the status board

3 Measured behaviour

A run of 8 samples through 6 positions in the simulation (535 s simulated, filtration time-compressed to 70 to 110 s per sample). 8 of 8 wet transfers were laid flat and centred, with the arm running at twice the speed of the first study (re-checked on 48 randomised transfers: 96% success). Station times measured in the run:

Step	Mean (s)	Longest (s)	Runs
Arm: dry membrane from magazine to frit	17	18	8
Arm: wet membrane from frit to agar	15	15	8
Arm: tip wash	5	5	8
Doser: tip, aspirate, dispense, eject	34	65	8
Shuttle: plate in, lid off	14	14	8
Shuttle: lid on, plate out	16	16	8
Position: clean-in-place	38	38	8

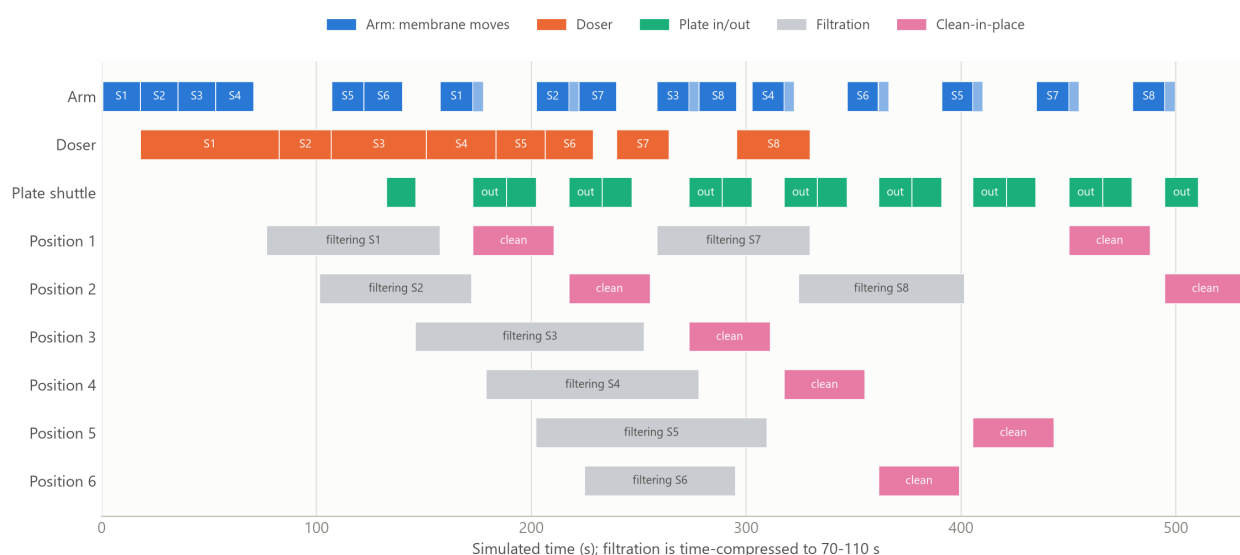


Figure 2. Timeline of the run. The arm (top lane) is the only shared resource; doser, shuttle and positions overlap with it.

Throughput. With these step times, real filtration times (median 180 s, with a 4% chance of a 10-minute stall) and a 2% transfer retry rate, a simulated 8-hour shift gives the following (mean of 5 runs):

Positions	Samples per hour	Minutes per sample	100 samples take	Arm busy	Doser busy
2	22	2.70	4.5 h	23%	21%
3	32	1.85	3.1 h	34%	30%
4	41	1.45	2.4 h	43%	39%
6	59	1.02	1.7 h	60%	55%
8	68	0.88	1.5 h	71%	64%
12	70	0.85	1.4 h	72%	66%

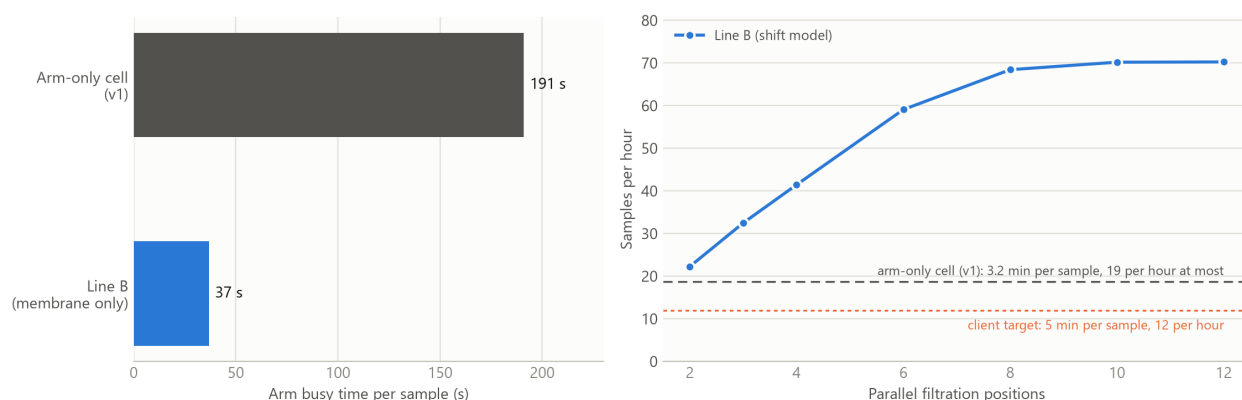


Figure 3. Left: arm busy time per sample. Right: shift-model throughput against positions, with the arm-only cell and the client target.

With slow filtrations (10 minutes, no variation) six positions still deliver 29 samples per hour. Throughput levels off at about 70 per hour beyond 8 positions: arm (71% busy) and doser (64%) both need the manifold zone, so they take turns, and each sample still needs its own arm moves. Further gains would come from separate access paths for doser and arm, or faster arm cycles. The lab's actual need, 100 samples per day, is about 4 samples per hour, so the cell would work in short bursts and sit idle most of the day: even 2 positions give 22 per hour, above the 12 per hour set by the 5-minute target. More positions buy headroom for peaks and slow filtrations, not routine throughput.

4 Continuous supply of materials

Every consumable has a buffer sized for at least one day at the lab's rate, and the cell stops accepting new samples before any buffer runs dry. The operator works through a hatch, so the enclosure stays closed during normal operation.

Consumable	Buffer	At 100 samples/day	At 300 samples/day	At full speed
Membranes (sterile, in magazine)	150	35 h	12 h	2.5 h
Agar plates (hotel)	100	24 h	8 h	1.7 h
Dosing tips (rack)	96	23 h	8 h	1.6 h
Sanitant and rinse water	see Section 5	96 h	32 h	6.8 h
Sample vessels (racks through the hatch)	barcode-scanned racks	continuous	continuous	continuous

- **In:** racks of septum-capped vessels, membrane magazines and plate stacks through the hatch; one vessel barcode is read at entry and linked to the plate barcode at exit.
- **Out:** closed plates stacked for the incubator (or into a hotel that docks to it); used tips to a sealed autoclave bag; waste water to drain.

- **Monitoring:** every buffer is counted; the scheduler refuses new samples below a threshold and raises an operator request, so the cell never starves mid-sample.

5 Disinfection of the tools

The three disinfection options discussed so far each have a drawback: ethanol needs explosion-proof extraction, hydrogen peroxide is costly, and UV leaves shadows. Line B reduces the problem to places where a liquid can reach everything, and avoids ethanol:

Surface	Concept	Duration in the simulation
Funnel and frit (sample contact)	clean-in-place on the position: sterile-water rinse, sanitant, rinse, air dry, funnel closed	38 s per position, parallel to the arm
Tweezer tips (membrane contact)	dip in a sanitant bath, then air knife, after every wet transfer	5 s, inside the arm cycle
Dosing	one disposable tip per sample; the nozzle never touches the sample	none
Plate and lid	closed in the hotel, opened only at the transfer spot	none

Choice of sanitant. The concept works with a non-flammable sanitant (hot water at 80 °C or above, peracetic acid, a quaternary ammonium or hypochlorous solution), which removes the ATEX question. Which one is acceptable depends on membrane and frit compatibility and on residue limits, to be settled with the lab.

Validation. A process blank is filtered at fixed intervals (sterile water through a cleaned position, then onto agar and incubated); a growing blank flags the position and triggers an extra cycle.

6 Variants and decisions

Variant	Idea	Strengths	Risks
B (this note)	fixed stations, arm only for the membrane	fast, fully standard Petri dishes and membranes, cleaning on the position	wet membrane handling still needs the arm; more mechanisms to build
C: cassette	single-use funnel with membrane in a rigid base that snaps onto a media cassette; no wet membrane handling	removes the hardest manipulation and most of the cleaning	proprietary consumables and per-sample cost; lab must accept cassettes instead of its own agar
A (v1)	arm does everything	proven in simulation, flexible	3.2 min per sample, arm-limited

Decisions for the lab. (1) Will standard Petri dishes with its own agar remain, or are cassettes acceptable? (2) Can vessels switch to septum caps? (3) Typical and worst-case filtration times and stall rates. (4) Cleaning standard between samples and the sanitant. (5) Should plates go straight into an incubator?

7 What the simulation does not settle, and next steps

- Wet-membrane behaviour (adhesion, surface tension, air entrapment) is not modelled; it is measured first on the real Franka.
- Tweezer tip design, funnel sealing and vacuum behaviour of the manifold, dosing accuracy and carry-over: hardware questions.
- Fixed stations are animated, not designed; they are the basis for a cost estimate and a concept sketch, not a mechanical design.
- Filtration is time-compressed in the run; the shift model uses real times and the step durations measured in the run.

Proposed next steps

- **Week 1:** real-membrane transfer tests on the Franka (Section 7 of the first report); price the doser, the lifting-funnel manifold and the plate shuttle.
- **Week 2:** choose sanitant and test clean-in-place on one funnel and frit with process blanks; decide Petri dish versus cassette.
- **Then:** detailed cell layout, risk register and budget for a pilot cell.